

THERMODYNAMIC FORMALISM MEETS ERGODIC OPTIMIZATION

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This note presents a concrete example illustrating the following connection between the thermodynamic formalism and ergodic optimization, in the context of MANNEVILLE–POMEAU maps and HÖLDER continuous potentials:

(*) *If a potential undergoes a phase transition in temperature, then the DIRAC measure at the neutral fixed point uniquely maximizes the potential.*

This result is a direct consequence of [CIR25, Key Lemma and Proposition 2.10] and the concrete example appears as [CIR25, Example 1.2]. For background on ergodic optimization, see, for example, [Jen19].

The example relies on the following notation and terminology. Fix α in $(0, +\infty)$ and the MANNEVILLE–POMEAU or *intermittent* map $f: [0, 1] \rightarrow [0, 1]$ defined by

$$(1) \quad f(x) := \begin{cases} x(1 + x^\alpha) & \text{if } x(1 + x^\alpha) \leq 1; \\ x(1 + x^\alpha) - 1 & \text{otherwise.} \end{cases}$$

Denote by $\mathcal{M}$ the space of BOREL probability measures on $[0, 1]$ that are invariant by f and, for each ν in $\mathcal{M}$, by h_ν the measure-theoretic entropy of ν . For each continuous function $\varphi: [0, 1] \rightarrow \mathbb{R}$, the *pressure* $P(\varphi)$ of f for the potential φ is defined by

$$(2) \quad P(\varphi) := \sup \left\{ h_\nu + \int \varphi \, d\nu : \nu \in \mathcal{M} \right\}.$$

Proposition 1. *Let γ in $(0, +\infty)$ be given, and define the potential $\hat{\varphi}$ by*

$$(3) \quad \hat{\varphi}(x) := -x^\gamma(1 - x).$$

Then, $t \mapsto P(t\hat{\varphi})$ is real-analytic on $(0, +\infty)$.

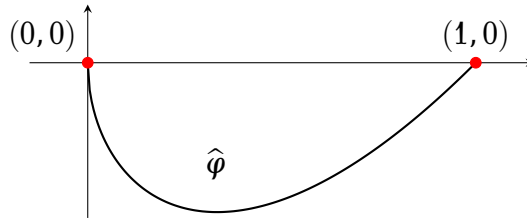


FIGURE 1. The graph of the potential $\hat{\varphi}$ with $\gamma = \frac{1}{2}$.

The potential $\hat{\varphi}$ is nonpositive and vanishes precisely at the neutral fixed point 0 and at the repelling fixed point 1. Thus, the measure δ_0 maximizes $\hat{\varphi}$, but not uniquely. The conclusion of **Proposition 1** is that the potential $\hat{\varphi}$ does not undergo a phase transition in temperature, as ensured by (*).

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The remainder of this note proves [Proposition 1](#). The proof relies on a few lemmas. Denote by x_1 the unique discontinuity point of f .

Lemma 2. *For every β in $(0, +\infty)$,*

$$(4) \quad \sup_{[0,1]} \beta \hat{\varphi} < P(\beta \hat{\varphi}).$$

In particular, the potential $\beta \hat{\varphi}$ is hyperbolic for f in the sense of [\[IRRL12\]](#).

Proof. Denote by x_2 the unique preimage of x_1 by f in $(0, x_1)$, and by X the maximal invariant set of f in $[x_2, 1]$. Then, $f|_X$ is uniformly expanding and topologically conjugate to the golden mean shift, via a HÖLDER continuous conjugacy. By the thermodynamic formalism for HÖLDER continuous potentials, it follows that $f|_X$ has a unique GIBBS state for the potential $\beta \hat{\varphi}|_X$ and that the measure-theoretic entropy of this measure is strictly positive. See, for example, [\[Bow08, Theorem 1.25\]](#). It follows that δ_1 is not an equilibrium state of $f|_X$ for the potential $\beta \hat{\varphi}|_X$, and thus

$$(5) \quad P(\beta \hat{\varphi}) \geq P_{f|_X}(\beta \hat{\varphi}|_X) > h_{\delta_1}(f) + \int \beta \hat{\varphi}|_X d\delta_1 = 0.$$

Since $\hat{\varphi} \leq 0$ on $[0, 1]$, the inequality [\(4\)](#) follows. $\square$

Lemma 3. *Let β be in $(0, +\infty)$ and put $g := \exp(\beta \hat{\varphi} - P(\beta \hat{\varphi}))$. Then, there is a $(1/g)$ -conformal probability measure m for f that is atom-free and satisfies $\text{supp}(m) = [0, 1]$.*

The proof of this lemma relies on the following one. Put $\Sigma := \{0, 1\}^{\mathbb{N}_0}$ and endow it with the distance dist defined, for distinct $(x_n)_{n=0}^{+\infty}$ and $(x'_n)_{n=0}^{+\infty}$ in Σ , by

$$(6) \quad \text{dist}((x_n)_{n=0}^{+\infty}, (x'_n)_{n=0}^{+\infty}) := 2^{-\min\{n: n \in \mathbb{N}_0, x_n \neq x'_n\}}.$$

The *shift map* $\sigma: \Sigma \rightarrow \Sigma$ is defined by $\sigma((x_n)_{n=0}^{+\infty}) := (x_{n+1})_{n=0}^{+\infty}$. Denote by $C(\Sigma)$ the BANACH space of real-valued continuous functions defined on Σ endowed with the supremum norm $\|\cdot\|_\Sigma$, and by $\mathbf{1}_\Sigma$ the function in $C(\Sigma)$ that is constant equal to 1. For ϕ in $C(\Sigma)$ denote by $P_\sigma(\phi)$ the pressure of σ for the potential ϕ . The *transfer operator* $\mathcal{L}_{\sigma, \phi}$ of σ for the potential ϕ is defined, for ψ in $C(\Sigma)$, by

$$(7) \quad \mathcal{L}_{\sigma, \phi} \psi(\underline{x}) := \sum_{\underline{x}' \in \sigma^{-1}(\underline{x})} \exp(\phi(\underline{x}')) \psi(\underline{x}').$$

Lemma 4. *For every ϕ in $C(\Sigma)$, the sequence $(\frac{1}{n} \log \mathcal{L}_{\sigma, \phi}^n \mathbf{1}_\Sigma)_{n=1}^{+\infty}$ converges uniformly to $P_\sigma(\phi) \cdot \mathbf{1}_\Sigma$.*

Proof. For all n in $\mathbb{N}$ and $x_0 \dots x_{n-1}$ in $\{0, 1\}$, the *cylinder* $[x_0 \dots x_{n-1}]$ is defined by

$$(8) \quad [x_0 \dots x_{n-1}] := \{(x'_n)_{n=0}^{+\infty} \in \Sigma: x'_m = x_m, m \in \{0, \dots, n-1\}\}.$$

The Variational Principle yields

$$(9) \quad \lim_{n \rightarrow +\infty} \frac{1}{n} \log \sum_{x_0, \dots, x_{n-1} \in \{0, 1\}} \sup_{\underline{x} \in [x_0 \dots x_{n-1}]} \exp(S_n(\phi)(\underline{x})) = P_\sigma(\phi),$$

see, for example, [\[Bow08, Theorem 2.17\]](#).

Let ε in $(0, +\infty)$ be given. By uniform continuity, there is n_0 in $\mathbb{N}$ such that, for all $x_0, \dots, x_{n_0-1}$ in $\{0, 1\}$ and $\underline{x}$ and $\underline{x}'$ in $[x_0 \dots x_{n_0-1}]$, the inequality $|\phi(\underline{x}) - \phi(\underline{x}')| \leq \varepsilon$ holds. It follows that, for all n in $\mathbb{N}$ satisfying $n \geq n_0$, $x_0, \dots, x_{n-1}$ in $\{0, 1\}$, and $\underline{x}$ and $\underline{x}'$ in $[x_0 \dots x_n]$,

$$(10) \quad |S_n(\phi)(\underline{x}) - S_n(\phi)(\underline{x}')| \leq (n - n_0)\varepsilon + 2n_0\|\phi\|_\Sigma.$$

Thus,

$$(11) \quad \mathcal{L}_{\sigma, \phi}^n \mathbf{1}_\Sigma(\underline{x}) \leq \sum_{x_0, \dots, x_{n-1} \in \{0, 1\}} \sup_{\underline{x}' \in [x_0 \dots x_{n-1}]} \exp(S_n(\phi)(\underline{x}')) \leq \exp((n - n_0)\varepsilon + 2n_0\|\phi\|_\Sigma) \mathcal{L}_{\sigma, \phi}^n \mathbf{1}_\Sigma(\underline{x})$$

and

$$(12) \quad \left| \frac{1}{n} \log \mathcal{L}_{\sigma, \phi}^n \mathbf{1}_\Sigma(\underline{x}) - \frac{1}{n} \log \sum_{x_0, \dots, x_{n-1} \in \{0, 1\}} \sup_{\underline{x}' \in [x_0 \dots x_{n-1}]} \exp(S_n(\phi)(\underline{x}')) \right| \leq \frac{n - n_0}{n} \varepsilon + \frac{2n_0}{n} \|\phi\|_\Sigma.$$

Together with (9), this yields the lemma. $\square$

Proof of Lemma 3. Put $I := [0, 1]$ and let $\tilde{f}: \tilde{I} \rightarrow \tilde{I}$ be the continuous extension of f obtained by “doubling” the points of $\bigcup_{n=0}^\infty f^{-n}(x_1)$, as in [CIR25, Proof of Lemma 2.6]. Put $Y^+ := \bigcup_{n=0}^{+\infty} \tilde{f}^{-n}(x_1^+)$, and denote by $\pi: \tilde{I} \rightarrow I$ the corresponding projection and $\tilde{\varphi}: \tilde{I} \rightarrow \mathbb{R}$ the continuous extension of $\hat{\varphi}|_{\tilde{I} \setminus Y^+}$. Then, $\pi \circ \tilde{f} = f \circ \pi$, $\tilde{\varphi} = \hat{\varphi} \circ \pi$, and $\pi: \tilde{I} \setminus Y^+ \rightarrow I$ is a BOREL measurable bijection conjugating $\tilde{f}|_{\tilde{I} \setminus Y^+}$ to f .

Since $\tilde{f}$ is topologically conjugate to the full shift in two symbols [CIR25, Lemma 2.6], the dual $\mathcal{L}_{\tilde{f}, \tilde{\varphi}}^*$ of the transfer operator $\mathcal{L}_{\tilde{f}, \tilde{\varphi}}$ of $\tilde{f}$ for the potential $\tilde{\varphi}$ is continuous and has an eigenprobability $\tilde{m}$ by the SCHAUDER–TYCHONOFF Fixed Point Theorem, see, for example, [Bow08, p.9]. Denoting by λ the eigenvalue of $\tilde{m}$, it follows that, for every n in $\mathbb{N}$,

$$(13) \quad \lambda^n \tilde{m}(\tilde{I}) = (\mathcal{L}_{\tilde{f}, \tilde{\varphi}}^*)^n \tilde{m}(\tilde{I}) = \int \mathbf{1}_{\tilde{I}} d(\mathcal{L}_{\tilde{f}, \tilde{\varphi}}^*)^n \tilde{m} = \int \mathcal{L}_{\tilde{f}, \tilde{\varphi}}^n \mathbf{1}_{\tilde{I}} d\tilde{m}.$$

Combined with Lemma 4, this yields

$$(14) \quad \log \lambda = \lim_{n \rightarrow +\infty} \frac{1}{n} \log \int \mathcal{L}_{\tilde{f}, \tilde{\varphi}}^n \mathbf{1}_{\tilde{I}} d\tilde{m} = P_{\tilde{f}}(\tilde{\varphi})$$

and thus $\mathcal{L}_{\tilde{f}, \tilde{\varphi}}^* \tilde{m} = \exp(P_{\tilde{f}}(\tilde{\varphi})) \tilde{m}$. It follows that $\tilde{m}$ is $\exp(P(\beta\tilde{\varphi}) - \beta\tilde{\varphi})$ -conformal for $\tilde{f}$. Since $\tilde{f}$ is topologically exact on $\tilde{I}$, the equality $\text{supp}(\tilde{m}) = \tilde{I}$ holds.

To prove that $\tilde{m}$ is atom-free, let $\tilde{x}$ in $\tilde{I}$ be given. For every k in $\mathbb{N}$,

$$(15) \quad \tilde{m}(\{x\}) = \exp(-(kP(\hat{\varphi}) - S_k(\tilde{\varphi})(\tilde{x}))) \tilde{m}(\{\tilde{f}^k(x)\}) \leq \exp\left(-k \left(P(\hat{\varphi}) - \sup_{[0, 1]} \hat{\varphi}\right)\right) \tilde{m}(\{\tilde{f}^k(\tilde{x})\}).$$

Letting $k \rightarrow +\infty$ and using (4) in Lemma 2 yields $\tilde{m}(\{\tilde{x}\}) = 0$. This proves that $\tilde{m}$ is atom-free.

Put $m := \pi_* \tilde{m}$. Since $\tilde{m}$ is atom-free and $\exp(P(\beta\tilde{\varphi}) - \beta\tilde{\varphi})$ -conformal for $\tilde{f}$, the measure m is atom-free and $(1/g)$ -conformal for f . Moreover, $\tilde{m}(\tilde{I}) = 1$ and $\text{supp}(\tilde{m}) = \tilde{I}$ imply $m(I) = 1$ and $\text{supp}(m) = I$. $\square$

Proof of Proposition 1. Let β be in $(0, +\infty)$, put $g := \exp(\beta\hat{\varphi} - P(\beta\hat{\varphi}))$ as in Lemma 3, and put $\gamma_* := \min\{1, \gamma\}$. The functions $\hat{\varphi}$ and g belong to $C^{\gamma_*}([0, 1], \mathbb{R})$ and g satisfies $\sup_{[0, 1]} g < 1$ by Lemma 2. Let m be the atom-free $(1/g)$ -conformal probability measure for f satisfying $\text{supp}(m) = [0, 1]$ given by Lemma 3.

The first step is to verify the hypotheses of [LRL14, Theorem 1 in §4.2], which is a reformulation of results of KELLER [Kel85, Theorems 3.2 and 3.3]. Hypotheses H1 and H2 of [LRL14, Theorem 1 in §4.2] follow automatically from the fact that every function in $C^{\gamma_*}([0, 1], \mathbb{R})$ is of bounded $(1/\gamma_*)$ -variation and that m is $(1/g)$ -conformal for f . Hypothesis H3 of [LRL14, Theorem 1 in §4.2] can be shown as in the proof of [LRL14, Lemma 5.1]. Thus, the hypotheses of [LRL14, Theorem 1 in §4.2] are satisfied. Combined with the fact that f is topologically exact on $(0, 1]$ by [CIR25, Lemma 2.8] and well-known arguments, this result implies that $t \mapsto P(t\varphi)$ is real-analytic at $t = \beta$. $\square$

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